SE LAB 7: Fuzzing for Input Validation Bugs

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Screenshots:

1. Original Application Execution

```
dhritirajesh@M1-R4VM2F39L9 L6_L7_58_restaurant_bill % cd LAB7
python3 restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *paneer tikka*
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,180,320
Item prices: [250, 180, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/4
Each person pays: 300
dhritirajesh@M1-R4VM2F39L9 LAB7 %
```

2. Fuzz Testing – Failing Test Cases

```
dhritirajesh@M1-R4VM2F39L9 LAB7 % python3 -m pytest test_restaurant_bill_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/dhritirajesh/L6_L7_58_restaurant_bill/LAB7
plugins: hypothesis-6.141.1, Faker-37.12.0
collected 3 items

test_restaurant_bill_parser.py FFF                                       [100%]

=================================== FAILURES ===================================
_____________________________ test_clean_dish_name_no_crash _____________________________

s = None

    @given(st.text() | st.none())
    def test_clean_dish_name_no_crash(s):
        try:
>           clean_dish_name(s)

test_restaurant_bill_parser.py:9:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

name = None

    def clean_dish_name(name):
        """
        Removes the chef's-special markers (*, #, ~) and title-cases the name.
        Assumes name is a non-empty string.
        """
>       name = name.strip()
E       AttributeError: 'NoneType' object has no attribute 'strip'

restaurant_bill_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_clean_dish_name_no_crash(s):

test_restaurant_bill_parser.py:7:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None
```

3. Fixed Application Execution

```
dhritirajesh@M1-R4VM2F39L9 LAB7 % python3 restaurant_bill_parser.py
==== Running clean_dish_name ====
Enter a dish name from the menu (e.g. *paneer tikka*): *paneer tikka*
Dish name: Paneer Tikka

==== Running parse_item_prices ====
Enter item prices separated by commas (e.g. 250,180,320): 250,180,320
Item prices: [250, 180, 320]

==== Running split_amount ====
Enter the bill and number of people as TOTAL/PEOPLE (e.g. 1200/4): 1200/4
Each person pays: 300
dhritirajesh@M1-R4VM2F39L9 LAB7 %
```

4. Fuzz Testing – All Test Cases Passed

```
dhritirajesh@M1-R4VM2F39L9 LAB7 % python3 -m pytest test_restaurant_bill_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/dhritirajesh/L6_L7_58_restaurant_bill/LAB7
plugins: hypothesis-6.141.1, Faker-37.12.0
collected 3 items

test_restaurant_bill_parser.py ...                                       [100%]

============================== 3 passed in 0.36s ===============================
dhritirajesh@M1-R4VM2F39L9 LAB7 %
```